

Ten Common Proof Questions for Regression

Midterm Practice with Complete Solutions

STAT452 – Applied Statistics for Engineers and Scientists II

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Setup and notation

Unless a question states otherwise, use the simple linear regression model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n,$$

where the predictor values x_i are fixed, $E(\varepsilon_i) = 0$, $\text{Var}(\varepsilon_i) = \sigma^2$, and the errors are uncorrelated. Define

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i,$$

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2, \quad S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}), \quad S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2.$$

We assume $S_{xx} > 0$, so not all predictor values are equal. The fitted values and residuals are

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i, \quad \hat{e}_i = y_i - \hat{y}_i.$$

Exam strategy. For most proofs, first write the model and the relevant estimator in a centered form. Then use one of the key identities

$$\sum_{i=1}^n (x_i - \bar{x}) = 0, \quad \sum_{i=1}^n (y_i - \bar{y}) = 0, \quad \sum_{i=1}^n \hat{e}_i = \sum_{i=1}^n x_i \hat{e}_i = 0.$$

1 Derive the least-squares estimators

Question 1. Starting from the residual sum of squares

$$Q(b_0, b_1) = \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2,$$

prove that its unique minimizers are

$$\boxed{\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}}, \quad \boxed{\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}}.$$

Solution. Differentiate Q with respect to the two unknown coefficients:

$$\begin{aligned} \frac{\partial Q}{\partial b_0} &= -2 \sum_{i=1}^n (y_i - b_0 - b_1 x_i), \\ \frac{\partial Q}{\partial b_1} &= -2 \sum_{i=1}^n x_i (y_i - b_0 - b_1 x_i). \end{aligned}$$

Setting both derivatives equal to zero gives the normal equations

$$\sum_{i=1}^n y_i = nb_0 + b_1 \sum_{i=1}^n x_i, \tag{1}$$

$$\sum_{i=1}^n x_i y_i = b_0 \sum_{i=1}^n x_i + b_1 \sum_{i=1}^n x_i^2. \tag{2}$$

Divide (1) by n :

$$\bar{y} = b_0 + b_1 \bar{x},$$

so

$$b_0 = \bar{y} - b_1 \bar{x}.$$

Substitute this expression into (2):

$$\begin{aligned} \sum x_i y_i &= (\bar{y} - b_1 \bar{x}) \sum x_i + b_1 \sum x_i^2 \\ &= n\bar{x}\bar{y} + b_1 \left(\sum x_i^2 - n\bar{x}^2 \right). \end{aligned}$$

Therefore

$$b_1 = \frac{\sum x_i y_i - n\bar{x}\bar{y}}{\sum x_i^2 - n\bar{x}^2}.$$

The numerator and denominator are exactly the centered sums

$$\sum x_i y_i - n\bar{x}\bar{y} = S_{xy}, \quad \sum x_i^2 - n\bar{x}^2 = S_{xx}.$$

Hence $b_1 = S_{xy}/S_{xx}$ and $b_0 = \bar{y} - b_1 \bar{x}$.

It remains to verify that the stationary point is the unique minimum. The Hessian matrix is

$$2 \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}.$$

Its first leading principal minor is $2n > 0$, and its determinant is

$$4 \left(n \sum x_i^2 - \left(\sum x_i \right)^2 \right) = 4nS_{xx} > 0.$$

Thus the Hessian is positive definite, so the stationary point is the unique global minimizer. $\square$

2 Prove equivalent formulas for the slope

Question 2. Prove the computational identities

$$S_{xx} = \sum x_i^2 - \frac{(\sum x_i)^2}{n}, \quad S_{xy} = \sum x_i y_i - \frac{(\sum x_i)(\sum y_i)}{n},$$

and then prove

$$\boxed{\hat{\beta}_1 = r \frac{s_y}{s_x}},$$

where r is the sample correlation and s_x, s_y are the sample standard deviations.

Solution. Expand the definition of S_{xx} :

$$\begin{aligned} S_{xx} &= \sum (x_i - \bar{x})^2 \\ &= \sum x_i^2 - 2\bar{x} \sum x_i + n\bar{x}^2 \\ &= \sum x_i^2 - 2n\bar{x}^2 + n\bar{x}^2 \\ &= \sum x_i^2 - n\bar{x}^2 = \sum x_i^2 - \frac{(\sum x_i)^2}{n}. \end{aligned}$$

Similarly,

$$\begin{aligned} S_{xy} &= \sum (x_i - \bar{x})(y_i - \bar{y}) \\ &= \sum x_i y_i - \bar{y} \sum x_i - \bar{x} \sum y_i + n\bar{x}\bar{y} \\ &= \sum x_i y_i - n\bar{x}\bar{y} \\ &= \sum x_i y_i - \frac{(\sum x_i)(\sum y_i)}{n}. \end{aligned}$$

Now use

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}, \quad s_x = \sqrt{\frac{S_{xx}}{n-1}}, \quad s_y = \sqrt{\frac{S_{yy}}{n-1}}.$$

Then

$$\begin{aligned} r \frac{s_y}{s_x} &= \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} \sqrt{\frac{S_{yy}/(n-1)}{S_{xx}/(n-1)}} \\ &= \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} \sqrt{\frac{S_{yy}}{S_{xx}}} = \frac{S_{xy}}{S_{xx}} = \hat{\beta}_1. \end{aligned}$$

Thus the least-squares slope equals the correlation multiplied by the ratio of the sample standard deviations. $\square$

3 Prove the fundamental residual identities

Question 3. For a least-squares line containing an intercept, prove that

$$\boxed{\sum_{i=1}^n \hat{e}_i = 0}, \quad \boxed{\sum_{i=1}^n x_i \hat{e}_i = 0}, \quad \boxed{\sum_{i=1}^n \hat{y}_i = \sum_{i=1}^n y_i}.$$

Deduce that $\bar{\hat{y}} = \bar{y}$ and that the fitted line passes through $(\bar{x}, \bar{y})$.

Solution. At the least-squares solution, the two normal equations obtained by differentiating the residual sum of squares are

$$\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0 \quad \text{and} \quad \sum_{i=1}^n x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0.$$

Because $\hat{e}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$, these equations are precisely

$$\sum_{i=1}^n \hat{e}_i = 0, \quad \sum_{i=1}^n x_i \hat{e}_i = 0.$$

Also $y_i = \hat{y}_i + \hat{e}_i$, so summing over i gives

$$\sum y_i = \sum \hat{y}_i + \sum \hat{e}_i = \sum \hat{y}_i.$$

Dividing by n yields $\bar{\hat{y}} = \bar{y}$. Finally, evaluate the fitted line at $x = \bar{x}$:

$$\hat{y}(\bar{x}) = \hat{\beta}_0 + \hat{\beta}_1 \bar{x} = (\bar{y} - \hat{\beta}_1 \bar{x}) + \hat{\beta}_1 \bar{x} = \bar{y}.$$

Therefore the least-squares line passes through the data centroid $(\bar{x}, \bar{y})$. □

4 Prove the regression sum-of-squares decomposition

Question 4. Define

$$\text{SST} = \sum (y_i - \bar{y})^2, \quad \text{SSR} = \sum (\hat{y}_i - \bar{y})^2, \quad \text{SSE} = \sum (y_i - \hat{y}_i)^2.$$

Prove the ANOVA identity

$$\boxed{\text{SST} = \text{SSR} + \text{SSE}}.$$

State clearly why the cross-product term vanishes.

Solution. For each observation, add and subtract the fitted value:

$$y_i - \bar{y} = (\hat{y}_i - \bar{y}) + (y_i - \hat{y}_i) = (\hat{y}_i - \bar{y}) + \hat{e}_i.$$

Square and sum:

$$\sum (y_i - \bar{y})^2 = \sum (\hat{y}_i - \bar{y})^2 + \sum \hat{e}_i^2 + 2 \sum (\hat{y}_i - \bar{y}) \hat{e}_i.$$

The cross-product term is zero because

$$\begin{aligned}\sum (\hat{y}_i - \bar{y})\hat{e}_i &= \sum (\hat{\beta}_0 + \hat{\beta}_1 x_i - \bar{y})\hat{e}_i \\ &= (\hat{\beta}_0 - \bar{y}) \sum \hat{e}_i + \hat{\beta}_1 \sum x_i \hat{e}_i \\ &= 0,\end{aligned}$$

where the last line uses both normal-equation identities from Question 3. Consequently,

$$\text{SST} = \text{SSR} + \text{SSE}.$$

Geometrically, the centered fitted-value vector is orthogonal to the residual vector, so this is the Pythagorean theorem for the three sums of squares. $\square$

5 Prove unbiasedness of the coefficient estimators

Question 5. Under the fixed- x simple linear regression model, prove that

$$\boxed{\text{E}(\hat{\beta}_1) = \beta_1}, \quad \boxed{\text{E}(\hat{\beta}_0) = \beta_0}.$$

Solution. Substitute $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ into the centered slope formula. Since

$$Y_i - \bar{Y} = \beta_1(x_i - \bar{x}) + (\varepsilon_i - \bar{\varepsilon}),$$

we have

$$\begin{aligned}\hat{\beta}_1 &= \frac{\sum (x_i - \bar{x})(Y_i - \bar{Y})}{S_{xx}} \\ &= \frac{\beta_1 \sum (x_i - \bar{x})^2 + \sum (x_i - \bar{x})(\varepsilon_i - \bar{\varepsilon})}{S_{xx}}.\end{aligned}$$

Because $\sum (x_i - \bar{x}) = 0$, the term containing $\bar{\varepsilon}$ vanishes. Therefore

$$\hat{\beta}_1 = \beta_1 + \frac{\sum (x_i - \bar{x})\varepsilon_i}{S_{xx}}.$$

Taking expectation conditional on the fixed x_i values,

$$\text{E}(\hat{\beta}_1) = \beta_1 + \frac{\sum (x_i - \bar{x}) \text{E}(\varepsilon_i)}{S_{xx}} = \beta_1.$$

For the intercept,

$$\begin{aligned}\hat{\beta}_0 &= \bar{Y} - \hat{\beta}_1 \bar{x} \\ &= \beta_0 + \beta_1 \bar{x} + \bar{\varepsilon} - \hat{\beta}_1 \bar{x} \\ &= \beta_0 + \bar{\varepsilon} - \bar{x}(\hat{\beta}_1 - \beta_1).\end{aligned}$$

Thus

$$\text{E}(\hat{\beta}_0) = \beta_0 + \text{E}(\bar{\varepsilon}) - \bar{x}\{\text{E}(\hat{\beta}_1) - \beta_1\} = \beta_0.$$

Normality is not required for either unbiasedness result; zero-mean errors are sufficient. $\square$

6 Derive variances, covariance, and fitted-mean variance

Question 6. Assuming uncorrelated errors with common variance σ^2 , prove

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}},$$

$$\text{Var}(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right), \quad \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\bar{x}\sigma^2}{S_{xx}}.$$

Then prove that, at a fixed x_0 ,

$$\text{Var}\{\hat{\mu}(x_0)\} = \sigma^2 \left[\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right],$$

where $\hat{\mu}(x_0) = \hat{\beta}_0 + \hat{\beta}_1 x_0$.

Solution. From Question 5,

$$\hat{\beta}_1 - \beta_1 = \sum_{i=1}^n a_i \varepsilon_i, \quad a_i = \frac{x_i - \bar{x}}{S_{xx}}.$$

Using uncorrelated errors with variance σ^2 ,

$$\text{Var}(\hat{\beta}_1) = \sigma^2 \sum a_i^2 = \sigma^2 \frac{\sum (x_i - \bar{x})^2}{S_{xx}^2} = \frac{\sigma^2}{S_{xx}}.$$

Also

$$\hat{\beta}_0 - \beta_0 = \bar{\varepsilon} - \bar{x}(\hat{\beta}_1 - \beta_1).$$

We first calculate

$$\begin{aligned} \text{Cov}(\bar{\varepsilon}, \hat{\beta}_1 - \beta_1) &= \text{Cov}\left(\frac{1}{n} \sum \varepsilon_i, \sum a_i \varepsilon_i\right) \\ &= \frac{\sigma^2}{n} \sum a_i = \frac{\sigma^2}{n S_{xx}} \sum (x_i - \bar{x}) = 0. \end{aligned}$$

Hence

$$\begin{aligned} \text{Var}(\hat{\beta}_0) &= \text{Var}(\bar{\varepsilon}) + \bar{x}^2 \text{Var}(\hat{\beta}_1) - 2\bar{x} \text{Cov}(\bar{\varepsilon}, \hat{\beta}_1) \\ &= \frac{\sigma^2}{n} + \bar{x}^2 \frac{\sigma^2}{S_{xx}}. \end{aligned}$$

Furthermore,

$$\begin{aligned} \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) &= \text{Cov}\{\bar{\varepsilon} - \bar{x}(\hat{\beta}_1 - \beta_1), \hat{\beta}_1\} \\ &= 0 - \bar{x} \text{Var}(\hat{\beta}_1) = -\frac{\bar{x}\sigma^2}{S_{xx}}. \end{aligned}$$

For the fitted mean, it is helpful to rewrite the estimation error in centered form:

$$\hat{\mu}(x_0) - (\beta_0 + \beta_1 x_0) = \bar{\varepsilon} + (x_0 - \bar{x})(\hat{\beta}_1 - \beta_1).$$

The two terms on the right are uncorrelated, as shown above. Thus

$$\begin{aligned}\text{Var}\{\hat{\mu}(x_0)\} &= \text{Var}(\bar{\varepsilon}) + (x_0 - \bar{x})^2 \text{Var}(\hat{\beta}_1) \\ &= \sigma^2 \left[\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right].\end{aligned}$$

For a new individual response Y_0 , independent of the fitted sample, the prediction error also contains the new error ε_0 . Therefore its variance is the displayed expression plus σ^2 . $\square$

7 Prove that MSE is unbiased for the error variance

Question 7. In simple linear regression, prove

$$\boxed{E(\text{SSE}) = (n-2)\sigma^2} \quad \text{and hence} \quad \boxed{E(\text{MSE}) = \sigma^2},$$

where $\text{MSE} = \text{SSE}/(n-2)$. You may use matrix notation.

Solution. Let

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbf{X} = \begin{pmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix},$$

where $\mathbf{X}$ has rank 2. Define the hat matrix and residual-maker matrix

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top, \quad \mathbf{M} = \mathbf{I} - \mathbf{H}.$$

Because $\mathbf{H}\mathbf{X} = \mathbf{X}$, the residual vector is

$$\begin{aligned}\mathbf{e} &= \mathbf{Y} - \hat{\mathbf{Y}} = \mathbf{M}\mathbf{Y} \\ &= \mathbf{M}(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \mathbf{M}\boldsymbol{\varepsilon}.\end{aligned}$$

Both $\mathbf{H}$ and $\mathbf{M}$ are symmetric and idempotent. Therefore

$$\text{SSE} = \mathbf{e}^\top \mathbf{e} = \boldsymbol{\varepsilon}^\top \mathbf{M}^\top \mathbf{M} \boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}^\top \mathbf{M} \boldsymbol{\varepsilon}.$$

For any fixed matrix $\mathbf{A}$ and random vector with mean zero and covariance $\sigma^2 \mathbf{I}$,

$$E(\boldsymbol{\varepsilon}^\top \mathbf{A} \boldsymbol{\varepsilon}) = \sigma^2 \text{tr}(\mathbf{A}).$$

To see this, expand the quadratic form. All off-diagonal expectations are zero, while each diagonal expectation is $E(\varepsilon_i^2) = \sigma^2$. Thus

$$E(\text{SSE}) = \sigma^2 \text{tr}(\mathbf{M}) = \sigma^2 \{\text{tr}(\mathbf{I}) - \text{tr}(\mathbf{H})\}.$$

Since $\mathbf{H}$ is the projection onto the two-dimensional column space of $\mathbf{X}$, its rank is 2. An idempotent matrix has eigenvalues only 0 and 1, so its trace equals its rank. Consequently,

$$\text{tr}(\mathbf{M}) = n - 2 \implies E(\text{SSE}) = (n - 2)\sigma^2.$$

Finally,

$$E(\text{MSE}) = E\left(\frac{\text{SSE}}{n-2}\right) = \sigma^2.$$

The loss of two degrees of freedom occurs because two coefficients, β_0 and β_1 , are estimated from the data. $\square$

8 Prove the Gauss–Markov result for the slope

Question 8. Let $\tilde{\beta}_1 = \sum_{i=1}^n c_i Y_i$ be any linear unbiased estimator of β_1 . Prove that

$$\text{Var}(\tilde{\beta}_1) \geq \text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}.$$

Thus prove that the least-squares slope is BLUE: the best linear unbiased estimator of β_1 .

Solution. First determine the conditions required for unbiasedness. Since

$$\begin{aligned} E(\tilde{\beta}_1) &= \sum c_i(\beta_0 + \beta_1 x_i) \\ &= \beta_0 \sum c_i + \beta_1 \sum c_i x_i, \end{aligned}$$

this equals β_1 for every β_0, β_1 exactly when

$$\sum c_i = 0, \quad \sum c_i x_i = 1.$$

The OLS slope is a linear estimator with weights

$$a_i = \frac{x_i - \bar{x}}{S_{xx}}, \quad \hat{\beta}_1 = \sum a_i Y_i.$$

These weights satisfy

$$\sum a_i = 0, \quad \sum a_i x_i = \frac{\sum (x_i - \bar{x})x_i}{S_{xx}} = 1.$$

For any other linear unbiased estimator, write $c_i = a_i + d_i$. The two unbiasedness conditions imply

$$\sum d_i = 0, \quad \sum d_i x_i = 0.$$

The OLS weights are orthogonal to the perturbation weights because

$$\sum a_i d_i = \frac{1}{S_{xx}} \sum (x_i - \bar{x})d_i = \frac{1}{S_{xx}} \left(\sum x_i d_i - \bar{x} \sum d_i \right) = 0.$$

Using the common error variance and lack of correlation,

$$\begin{aligned} \text{Var}(\tilde{\beta}_1) &= \sigma^2 \sum c_i^2 \\ &= \sigma^2 \sum (a_i + d_i)^2 \\ &= \sigma^2 \sum a_i^2 + 2\sigma^2 \sum a_i d_i + \sigma^2 \sum d_i^2 \\ &= \text{Var}(\hat{\beta}_1) + \sigma^2 \sum d_i^2 \\ &\geq \text{Var}(\hat{\beta}_1). \end{aligned}$$

Equality holds only when all $d_i = 0$, meaning the competing estimator is the OLS estimator itself. This proof does not require normally distributed errors. $\square$

9 Prove R-squared equals squared correlation and F equals t-squared

Question 9. In simple linear regression with an intercept, prove

$$\boxed{R^2 = r_{xy}^2}.$$

Also prove that the overall regression F statistic for testing $H_0 : \beta_1 = 0$ is the square of the usual slope t statistic:

$$\boxed{F = t^2}.$$

Solution. Because the fitted line passes through $(\bar{x}, \bar{y})$,

$$\hat{y}_i - \bar{y} = \hat{\beta}_0 + \hat{\beta}_1 x_i - \bar{y} = \hat{\beta}_1 (x_i - \bar{x}).$$

Therefore

$$\begin{aligned} \text{SSR} &= \sum (\hat{y}_i - \bar{y})^2 = \hat{\beta}_1^2 \sum (x_i - \bar{x})^2 \\ &= \left(\frac{S_{xy}}{S_{xx}} \right)^2 S_{xx} = \frac{S_{xy}^2}{S_{xx}}. \end{aligned}$$

Since $\text{SST} = S_{yy}$,

$$R^2 = \frac{\text{SSR}}{\text{SST}} = \frac{S_{xy}^2}{S_{xx} S_{yy}} = \left(\frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}} \right)^2 = r_{xy}^2.$$

For the test of $H_0 : \beta_1 = 0$, the estimated standard error of the slope is

$$\text{se}(\hat{\beta}_1) = \sqrt{\frac{\text{MSE}}{S_{xx}}}.$$

Thus

$$t^2 = \left(\frac{\hat{\beta}_1}{\sqrt{\text{MSE}/S_{xx}}} \right)^2 = \frac{\hat{\beta}_1^2 S_{xx}}{\text{MSE}} = \frac{\text{SSR}}{\text{MSE}}.$$

The regression has one numerator degree of freedom, so its overall statistic is

$$F = \frac{\text{MSR}}{\text{MSE}} = \frac{\text{SSR}/1}{\text{MSE}} = \frac{\text{SSR}}{\text{MSE}}.$$

Hence $F = t^2$. The tests therefore give identical p -values for this two-sided slope hypothesis. $\square$

10 Derive the multiple-regression matrix formulas

Question 10. Consider the multiple linear regression model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbf{E}(\boldsymbol{\varepsilon}) = \mathbf{0}, \quad \text{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n,$$

where $\mathbf{X}$ is an $n \times p$ matrix of full column rank. Prove

$$\boxed{\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}},$$

$$\boxed{E(\hat{\beta}) = \beta}, \quad \boxed{\text{Var}(\hat{\beta}) = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}}.$$

Also prove that the hat matrix

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$$

is symmetric and idempotent.

Solution. The residual sum of squares is

$$S(\mathbf{b}) = (\mathbf{Y} - \mathbf{X}\mathbf{b})^\top (\mathbf{Y} - \mathbf{X}\mathbf{b}).$$

Expanding,

$$S(\mathbf{b}) = \mathbf{Y}^\top \mathbf{Y} - 2\mathbf{b}^\top \mathbf{X}^\top \mathbf{Y} + \mathbf{b}^\top \mathbf{X}^\top \mathbf{X} \mathbf{b}.$$

Differentiating with respect to $\mathbf{b}$ gives

$$\frac{\partial S}{\partial \mathbf{b}} = -2\mathbf{X}^\top \mathbf{Y} + 2\mathbf{X}^\top \mathbf{X} \mathbf{b}.$$

At the minimum,

$$\mathbf{X}^\top \mathbf{X} \hat{\beta} = \mathbf{X}^\top \mathbf{Y}.$$

Full column rank makes $\mathbf{X}^\top \mathbf{X}$ positive definite and invertible. Therefore

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

The Hessian is $2\mathbf{X}^\top \mathbf{X}$, which is positive definite, so this stationary point is the unique minimizer.

Substitute the model into the estimator:

$$\begin{aligned} \hat{\beta} &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\mathbf{X}\beta + \varepsilon) \\ &= \beta + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \varepsilon. \end{aligned}$$

Taking expectations gives

$$E(\hat{\beta}) = \beta.$$

For the covariance matrix, use $\text{Var}(\mathbf{AZ}) = \mathbf{A} \text{Var}(\mathbf{Z}) \mathbf{A}^\top$:

$$\begin{aligned} \text{Var}(\hat{\beta}) &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\sigma^2 \mathbf{I}) \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \\ &= \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \\ &= \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}. \end{aligned}$$

Finally,

$$\begin{aligned} \mathbf{H}^\top &= \left\{ \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \right\}^\top \\ &= \mathbf{X} \left\{ (\mathbf{X}^\top \mathbf{X})^{-1} \right\}^\top \mathbf{X}^\top = \mathbf{H}, \end{aligned}$$

because $\mathbf{X}^\top \mathbf{X}$ and its inverse are symmetric. Also

$$\begin{aligned} \mathbf{H}^2 &= \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \\ &= \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \mathbf{H}. \end{aligned}$$

Thus $\mathbf{H}$ is symmetric and idempotent; it is the orthogonal projection matrix onto the column space of $\mathbf{X}$. $\square$

One-page proof checklist

1. **OLS coefficients:** Differentiate SSE, obtain the normal equations, solve them, and mention positive definiteness for uniqueness.
2. **Centering trick:** Use $\sum(x_i - \bar{x}) = 0$ immediately.
3. **Residual orthogonality:** With an intercept, $\sum \hat{e}_i = 0$; with a predictor, $\sum x_i \hat{e}_i = 0$.
4. **ANOVA decomposition:** Expand $y_i - \bar{y} = (\hat{y}_i - \bar{y}) + \hat{e}_i$ and kill the cross term using the normal equations.
5. **Unbiasedness:** Write the estimator as the true parameter plus a weighted sum of zero-mean errors.
6. **Variance:** For uncorrelated equal-variance errors, $\text{Var}(\sum a_i \varepsilon_i) = \sigma^2 \sum a_i^2$.
7. **MSE:** Write $\text{SSE} = \boldsymbol{\varepsilon}^\top (\mathbf{I} - \mathbf{H}) \boldsymbol{\varepsilon}$ and use trace = residual degrees of freedom.
8. **Gauss–Markov:** Express any competing unbiased weights as OLS weights plus an orthogonal perturbation.
9. **Simple regression identities:** $\text{SSR} = \hat{\beta}_1^2 S_{xx} = S_{xy}^2 / S_{xx}$, which quickly gives both $R^2 = r^2$ and $F = t^2$.
10. **Matrix regression:** The normal equations are $\mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{Y}$.